
31.3 Modular arithmetic

Informally, you can think of modular arithmetic as arithmetic as usual over the integers, except that when working modulo n , then every result x is replaced by the element of $\{0, 1, \dots, n - 1\}$ that is equivalent to x , modulo n (so that x is replaced by $x \bmod n$). This informal model suffices if you stick to the operations of addition, subtraction, and multiplication. A more formal model for modular arithmetic, which follows, is best described within the framework of group theory.

Finite groups

A **group** $(S, \oplus)$ is a set S together with a binary operation $\oplus$ defined on S for which the following properties hold:

1. **Closure:** For all $a, b \in S$, we have $a \oplus b \in S$.
2. **Identity:** There exists an element $e \in S$, called the **identity** of the group, such that $e \oplus a = a \oplus e = a$ for all $a \in S$.
3. **Associativity:** For all $a, b, c \in S$, we have $(a \oplus b) \oplus c = a \oplus (b \oplus c)$.
4. **Inverses:** For each $a \in S$, there exists a unique element $b \in S$, called the **inverse** of a , such that $a \oplus b = b \oplus a = e$.

As an example, consider the familiar group $(\mathbb{Z}, +)$ of the integers $\mathbb{Z}$ under the operation of addition: 0 is the identity, and the inverse of a is $-a$. An **abelian group** $(S, \oplus)$ satisfies the **commutative law** $a \oplus b = b \oplus a$ for all $a, b \in S$. The **size** of group $(S, \oplus)$ is $|S|$, and if $|S| < \infty$, then $(S, \oplus)$ is a **finite group**.

The groups defined by modular addition and multiplication

We can form two finite abelian groups by using addition and multiplication modulo n , where n is a positive integer. These groups are based on the equivalence classes of the integers modulo n , defined in Section 31.1.

To define a group on $\mathbb{Z}_n$, we need suitable binary operations, which we obtain by redefining the ordinary operations of addition and multiplication. We can define addition and multiplication operations for $\mathbb{Z}_n$, because the equivalence class of two integers uniquely determines the equivalence class of their sum or product. That is, if $a = a' \pmod{n}$ and $b = b' \pmod{n}$, then

$$\begin{aligned} a + b &= a' + b' \pmod{n}, \\ ab &= a'b' \pmod{n}. \end{aligned}$$

Thus, we define addition and multiplication modulo n , denoted $+_n$ and $\cdot_n$, by

$$\begin{aligned} [a]_n +_n [b]_n &= [a + b]_n, \\ [a]_n \cdot_n [b]_n &= [ab]_n. \end{aligned} \tag{31.19}$$

(We can define subtraction similarly on $\mathbb{Z}_n$ by $[a]_n -_n [b]_n = [a - b]_n$, but division is more complicated, as we'll see.) These facts justify the common and convenient practice of using the smallest nonnegative element of each equivalence class as its representative when performing computations in $\mathbb{Z}_n$. We add, subtract, and multiply as usual on the representatives, but we replace each result x by the representative of its class, that is, by $x \bmod n$.

$+_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

(a)

$\cdot_{15}$	1	2	4	7	8	11	13	14
1	1	2	4	7	8	11	13	14
2	2	4	8	14	1	7	11	13
4	4	8	1	13	2	14	7	11
7	7	14	13	4	11	2	1	8
8	8	1	2	11	4	13	14	7
11	11	7	14	2	13	1	8	4
13	13	11	7	1	14	8	4	2
14	14	13	11	8	7	4	2	1

(b)

Figure 31.2 Two finite groups. Equivalence classes are denoted by their representative elements. **(a)** The group $(\mathbb{Z}_6, +_6)$. **(b)** The group $(\mathbb{Z}_{15}^*, \cdot_{15})$.

Using this definition of addition modulo n , we define the **additive group modulo n** as $(\mathbb{Z}_n, +_n)$. The size of the additive group modulo n is $|\mathbb{Z}_n| = n$. Figure 31.2(a) gives the operation table for the group $(\mathbb{Z}_6, +_6)$.

Theorem 31.12

The system $(\mathbb{Z}_n, +_n)$ is a finite abelian group.

Proof Equation (31.19) shows that $(\mathbb{Z}_n, +_n)$ is closed. Associativity and commutativity of $+_n$ follow from the associativity and commutativity of $+$:

$$\begin{aligned} ([a]_n +_n [b]_n) +_n [c]_n &= [a + b]_n +_n [c]_n \\ &= [(a + b) + c]_n \\ &= [a + (b + c)]_n \\ &= [a]_n +_n [b + c]_n \\ &= [a]_n +_n ([b]_n +_n [c]_n), \end{aligned}$$

$$\begin{aligned} [a]_n +_n [b]_n &= [a + b]_n \\ &= [b + a]_n \\ &= [b]_n +_n [a]_n. \end{aligned}$$

The identity element of $(\mathbb{Z}_n, +_n)$ is 0 (that is, $[0]_n$). The (additive) inverse of an element a (that is, of $[a]_n$) is the element $-a$ (that is, $[-a]_n$ or $[n - a]_n$), since $[a]_n +_n [-a]_n = [a - a]_n = [0]_n$. ■

Using the definition of multiplication modulo n , we define the **multiplicative group modulo n** as $(\mathbb{Z}_n^*, \cdot_n)$. The elements of this group are the set $\mathbb{Z}_n^*$ of elements in $\mathbb{Z}_n$ that are relatively prime to n , so that each one has a unique inverse, modulo n :

$$\mathbb{Z}_n^* = \{[a]_n \in \mathbb{Z}_n : \gcd(a, n) = 1\}.$$

To see that $\mathbb{Z}_n^*$ is well defined, note that for $0 \leq a < n$, we have $a = (a + kn) \pmod{n}$ for all integers k . By Exercise 31.2-3, therefore, $\gcd(a, n) = 1$ implies $\gcd(a + kn, n) = 1$ for all integers k . Since $[a]_n = \{a + kn : k \in \mathbb{Z}\}$, the set $\mathbb{Z}_n^*$ is well defined. An example of such a group is

$$\mathbb{Z}_{15}^* = \{1, 2, 4, 7, 8, 11, 13, 14\} ,$$

where the group operation is multiplication modulo 15. (We have denoted an element $[a]_{15}$ as a , and thus, for example, we denote $[7]_{15}$ as 7.) Figure 31.2(b) shows the group $(\mathbb{Z}_{15}^*, \cdot_{15})$. For example, $8 \cdot 11 = 13 \pmod{15}$, working in $\mathbb{Z}_{15}^*$. The identity for this group is 1.

Theorem 31.13

The system $(\mathbb{Z}_n^*, \cdot_n)$ is a finite abelian group.

Proof Theorem 31.6 implies that $(\mathbb{Z}_n^*, \cdot_n)$ is closed. Associativity and commutativity can be proved for $\cdot_n$ as they were for $+_n$ in the proof of Theorem 31.12. The identity element is $[1]_n$. To show the existence of inverses, let a be an element of $\mathbb{Z}_n^*$ and let (d, x, y) be returned by EXTENDED-EUCLID(a, n). Then we have $d = 1$, since $a \in \mathbb{Z}_n^*$, and

$$ax + ny = 1 , \tag{31.20}$$

or equivalently,

$$ax = 1 \pmod{n}.$$

Thus $[x]_n$ is a multiplicative inverse of $[a]_n$, modulo n . Furthermore, we claim that $[x]_n \in \mathbb{Z}_n^*$. To see why, equation (31.20) demonstrates that the smallest positive linear combination of x and n must be 1. Therefore, Theorem 31.2 implies that $\gcd(x, n) = 1$. We defer the proof that inverses are uniquely defined until Corollary 31.26 in Section 31.4. ■

As an example of computing multiplicative inverses, suppose that $a = 5$ and $n = 11$. Then EXTENDED-EUCLID(a, n) returns $(d, x, y) = (1, -2, 1)$, so that $1 = 5 \cdot (-2) + 11 \cdot 1$. Thus, $[-2]_{11}$ (i.e., $[9]_{11}$) is the multiplicative inverse of $[5]_{11}$.

When working with the groups $(\mathbb{Z}_n, +_n)$ and $(\mathbb{Z}_n^*, \cdot_n)$ in the remainder of this chapter, we follow the convenient practice of denoting equivalence classes by their representative elements and denoting the operations $+_n$ and $\cdot_n$ by the usual arithmetic notations $+$ and $\cdot$ (or juxtaposition, so that $ab = a \cdot b$) respectively. Furthermore, equivalences modulo n may also be interpreted as equations in $\mathbb{Z}_n$. For example, the following two statements are equivalent:

$$ax = b \pmod{n}$$

and

$$[a]_n \cdot_n [x]_n = [b]_n.$$

As a further convenience, we sometimes refer to a group $(S, \oplus)$ merely as S when the operation $\oplus$ is understood from context. We may thus refer to the groups $(\mathbb{Z}_n, +_n)$ and $(\mathbb{Z}_n^*, \cdot_n)$ as just $\mathbb{Z}_n$ and $\mathbb{Z}_n^*$, respectively.

We denote the (multiplicative) inverse of an element a by $(a^{-1} \pmod{n})$. Division in $\mathbb{Z}_n^*$ is defined by the equation $a/b = ab^{-1} \pmod{n}$. For example, in $\mathbb{Z}_{15}^*$ we have that $7^{-1} = 13 \pmod{15}$, since $7 \cdot 13 = 91 = 1 \pmod{15}$, so that $2/7 = 2 \cdot 13 = 11 \pmod{15}$.

The size of $\mathbb{Z}_n^*$ is denoted $\phi(n)$. This function, known as *Euler's phi function*, satisfies the equation

$$\phi(n) = n \prod_{p \text{ prime such that } p \mid n} \left(1 - \frac{1}{p}\right), \quad (31.21)$$

so that p runs over all the primes dividing n (including n itself, if n is prime). We won't prove this formula here. Intuitively, begin with a list of the n remainders $\{0, 1, \dots, n-1\}$ and then, for each prime p that divides n , cross out every multiple of p in the list. For example, since the prime divisors of 45 are 3 and 5,

$$\begin{aligned}
\phi(45) &= 45 \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) \\
&= 45 \left(\frac{2}{3}\right) \left(\frac{4}{5}\right) \\
&= 24.
\end{aligned}$$

If p is prime, then $\mathbb{Z}_p^* = \{1, 2, \dots, p-1\}$, and

$$\begin{aligned}
\phi(p) &= p \left(1 - \frac{1}{p}\right) \\
&= p - 1.
\end{aligned} \tag{31.22}$$

If n is composite, then $\phi(n) < n - 1$, although it can be shown that

$$\phi(n) > \frac{n}{e^\gamma \ln \ln n + 3 / \ln \ln n} \tag{31.23}$$

for $n \geq 3$, where $\gamma = 0.5772156649 \dots$ is *Euler's constant*. A somewhat simpler (but looser) lower bound for $n > 5$ is

$$\phi(n) > \frac{n}{6 \ln \ln n}. \tag{31.24}$$

The lower bound (31.23) is essentially the best possible, since

$$\liminf_{n \rightarrow \infty} \frac{\phi(n)}{n / \ln \ln n} = e^{-\gamma}. \tag{31.25}$$

Subgroups

If $(S, \oplus)$ is a group, $S' \subseteq S$, and $(S', \oplus)$ is also a group, then $(S', \oplus)$ is a *subgroup* of $(S, \oplus)$. For example, the even integers form a subgroup of the integers under the operation of addition. The following theorem, whose proof we leave as Exercise 31.3-3, provides a useful tool for recognizing subgroups.

Theorem 31.14 *(A nonempty closed subset of a finite group is a subgroup)*

If $(S, \oplus)$ is a finite group and S' is any nonempty subset of S such that $a \oplus b \in S'$ for all $a, b \in S'$, then $(S', \oplus)$ is a subgroup of $(S, \oplus)$.

■

For example, the set $\{0, 2, 4, 6\}$ forms a subgroup of $\mathbb{Z}_8$, since it is nonempty and closed under the operation $+$ (that is, it is closed under $+_8$).

The following theorem, whose proof is omitted, provides an extremely useful constraint on the size of a subgroup.

Theorem 31.15 (Lagrange's theorem)

If $(S, \oplus)$ is a finite group and $(S', \oplus)$ is a subgroup of $(S, \oplus)$, then $|S'|$ is a divisor of $|S|$.

■

A subgroup S' of a group S is a **proper** subgroup if $S' \neq S$. We'll use the following corollary in the analysis in Section 31.8 of the Miller-Rabin primality test procedure.

Corollary 31.16

If S' is a proper subgroup of a finite group S , then $|S'| \leq |S|/2$.

■

Subgroups generated by an element

Theorem 31.14 affords us a straightforward way to produce a subgroup of a finite group $(S, \oplus)$: choose an element a and take all elements that can be generated from a using the group operation. Specifically, define $a^{(k)}$ for $k \geq 1$ by

$$a^{(k)} = \bigoplus_{i=1}^k a = \underbrace{a \oplus a \oplus \cdots \oplus a}_k .$$

For example, taking $a = 2$ in the group $\mathbb{Z}_6$ yields the sequence

$$a^{(1)}, a^{(2)}, a^{(3)}, \dots = 2, 4, 0, 2, 4, 0, 2, 4, 0, \dots$$

We have $a^{(k)} = ka \bmod n$ in the group $\mathbb{Z}_n$, and $a^{(k)} = a^k \bmod n$ in the group $\mathbb{Z}_n^*$. We define the **subgroup generated by a** , denoted $\langle a \rangle$ or $(\langle a \rangle, \oplus)$, by

$$\langle a \rangle = \{a^{(k)} : k \geq 1\}.$$

We say that a **generates** the subgroup $\langle a \rangle$ or that a is a **generator** of $\langle a \rangle$. Since S is finite, $\langle a \rangle$ is a finite subset of S , possibly including all of S . Since the associativity of $\oplus$ implies

$$a^{(i)} \oplus a^{(j)} = a^{(i+j)},$$

$\langle a \rangle$ is closed and therefore, by Theorem 31.14, $\langle a \rangle$ is a subgroup of S . For example, in $\mathbb{Z}_6$, we have

$$\langle 0 \rangle = \{0\},$$

$$\langle 1 \rangle = \{0, 1, 2, 3, 4, 5\},$$

$$\langle 2 \rangle = \{0, 2, 4\}.$$

Similarly, in $\mathbb{Z}_7^*$, we have

$$\langle 1 \rangle = \{1\},$$

$$\langle 2 \rangle = \{1, 2, 4\},$$

$$\langle 3 \rangle = \{1, 2, 3, 4, 5, 6\}.$$

The **order** of a (in the group S), denoted $\text{ord}(a)$, is defined as the smallest positive integer t such that $a^{(t)} = e$. (Recall that $e \in S$ is the group identity.)

Theorem 31.17

For any finite group $(S, \oplus)$ and any $a \in S$, the order of a is equal to the size of the subgroup it generates, or $\text{ord}(a) = |\langle a \rangle|$.

Proof Let $t = \text{ord}(a)$. Since $a^{(t)} = e$ and $a^{(t+k)} = a^{(t)} \oplus a^{(k)} = a^{(k)}$ for $k \geq 1$, if $i > t$, then $a^{(i)} = a^{(j)}$ for some $j < i$. Therefore, as we generate elements by a , we see no new elements after $a^{(t)}$. Thus, $\langle a \rangle = \{a^{(1)}, a^{(2)}, \dots, a^{(t)}\}$, and so $|\langle a \rangle| \leq t$. To show that $|\langle a \rangle| \geq t$, we show that each element of the sequence $a^{(1)}, a^{(2)}, \dots, a^{(t)}$ is distinct. Suppose for the purpose of contradiction that $a^{(i)} = a^{(j)}$ for some i and j satisfying $1 \leq i$

$< j \leq t$. Then, $a^{(i+k)} = a^{(j+k)}$ for $k \geq 0$. But this equation implies that $a^{(i+(t-j))} = a^{(j+(t-j))} = e$, a contradiction, since $i + (t - j) < t$ but t is the least positive value such that $a^{(t)} = e$. Therefore, each element of the sequence $a^{(1)}, a^{(2)}, \dots, a^{(t)}$ is distinct, and $|\langle a \rangle| \geq t$. We conclude that $\text{ord}(a) = |\langle a \rangle|$. ■

Corollary 31.18

The sequence $a^{(1)}, a^{(2)}, \dots$ is periodic with period $t = \text{ord}(a)$, that is, $a^{(i)} = a^{(j)}$ if and only if $i = j \pmod{t}$. ■

Consistent with the above corollary, we define $a^{(0)}$ as e and $a^{(i)}$ as $a^{(i \pmod{t})}$, where $t = \text{ord}(a)$, for all integers i .

Corollary 31.19

If $(S, \oplus)$ is a finite group with identity e , then for all $a \in S$, $a^{(|S|)} = e$.

Proof Lagrange's theorem (Theorem 31.15) implies that $\text{ord}(a) \mid |S|$, and so $|S| = 0 \pmod{t}$, where $t = \text{ord}(a)$. Therefore, $a^{(|S|)} = a^{(0)} = e$. ■

Exercises

31.3-1

Draw the group operation tables for the groups $(\mathbb{Z}_4, +_4)$ and $(\mathbb{Z}_5^*, \cdot_5)$. Show that these groups are isomorphic by exhibiting a one-to-one correspondence f between $\mathbb{Z}_4$ and $\mathbb{Z}_5^*$ such that $a+b = c \pmod{4}$ if and only if $f(a) \cdot f(b) = f(c) \pmod{5}$.

31.3-2

List all subgroups of $\mathbb{Z}_9$ and of $\mathbb{Z}_{13}^*$.

31.3-3

Prove Theorem 31.14.

31.3-4

Show that if p is prime and e is a positive integer, then

$$\phi(p^e) = p^{e-1}(p - 1).$$

31.3-5

Show that for any integer $n > 1$ and for any $a \in \mathbb{Z}_n^*$, the function $f_a : \mathbb{Z}_n^* \rightarrow \mathbb{Z}_n^*$ defined by $f_a(x) = ax \bmod n$ is a permutation of $\mathbb{Z}_n^*$.

31.4 Solving modular linear equations

We now consider the problem of finding solutions to the equation

$$ax = b \pmod{n}, \tag{31.26}$$

where $a > 0$ and $n > 0$. This problem has several applications. For example, we'll use it in Section 31.7 as part of the procedure to find keys in the RSA public-key cryptosystem. We assume that a , b , and n are given, and we wish to find all values of x , modulo n , that satisfy equation (31.26). The equation may have zero, one, or more than one such solution.

Let $\langle a \rangle$ denote the subgroup of $\mathbb{Z}_n$ generated by a . Since $\langle a \rangle = \{a^{(x)} : x > 0\} = \{ax \bmod n : x > 0\}$, equation (31.26) has a solution if and only if $[b] \in \langle a \rangle$. Lagrange's theorem (Theorem 31.15) tells us that $|\langle a \rangle|$ must be a divisor of n . The following theorem gives us a precise characterization of $\langle a \rangle$.

Theorem 31.20

For any positive integers a and n , if $d = \gcd(a, n)$, then we have

$$\begin{aligned} \langle a \rangle &= \langle d \rangle \\ &= \{0, d, 2d, \dots, ((n/d) - 1)d\} \end{aligned}$$

in $\mathbb{Z}_n$, and thus